Natural Language Processing

Assignment- 3

TYPE OF QUESTION: MCQ

Number of questions: 7 **Total mark: 10 (Q5, Q6, Q7 carries two marks each)**

Question 1: Which of the following words contains both derivational and inflectional suffixes?

a) teacher
b) nationalization
c) kindness
d) disagreements

Answer: d

Solution:

Disagreements =
agree (root word) + ment (derivational suffix) + dis (derivational prefix) + s (inflectional suffix)

Question 2: Assume a fair coin is flipped three times in a row.
Let q be the probability of getting three heads consecutively.

Now consider a sentence of length M consisting of random binary digits (0s and 1s).
A language model assigns probability q to each digit in the sentence.

What is the perplexity of the sentence?

a) 4
b) 8
c) 16
d) 64

Answer: b

Solution:

Probability of head in one flip = 1/2

Probability of three heads in a row:
$q = (1/2)^3 = 1/8$

Probability of the entire sentence:
$P = (1/8)^M$

Perplexity:
$PP = P^{(-1/M)}$
$PP = ((1/8)^M)^{(-1/M)} = 8$

Question 3: Let x denote the input and y denote the class label.
Which of the following statements about generative and discriminative models are true?

1. Generative models implicitly define p(y | x) by learning the joint distribution p(x, y), but do not directly optimize the posterior during training.
2. Discriminative models require an explicit model of the input distribution p(x) in order to compute p(y | x).
3. Discriminative models directly optimize a conditional objective involving p(y | x).
4. Generative models learn a decision boundary without modeling class-conditional densities

a) 1 and 3
b) 1 and 4
c) 2 and 3
e) 3 and 4

Answer: a

Solution:

Statement 1 is correct. Generative models learn the joint distribution p(x, y). The posterior p(y | x) is obtained implicitly using Bayes' rule at inference time.

Statement 2 is incorrect. Discriminative models do not model the input distribution p(x).

Statement 3 is correct. Discriminative models directly optimize objectives based on the posterior p(y | x), such as cross-entropy loss.

Statement 4 is incorrect. Generative models explicitly model class-conditional distributions rather than only learning a decision boundary.

Question 4. Natural language processing is essentially the study of the meaning of the words a human says or writes. Natural language processing is all around us all the time, but it also happens to be a way to improve the chatbot or product we interact with on a regular basis. Natural language processing is all about mimicking our own language patterns. Natural language processing can also improve the efficiency of business transactions and customer care. Natural language processing is the application of computer technology.

Suppose we want to check the probabilities of the *final words* that succeed the *string* language processing in the above paragraph. Assume d= 0; it is also given that no of unigrams = 78, no of bigrams = 122, no of trigrams = 130,, Question 6 and Question 7 are related to Question 5 corpus.

Solve the question with the help of **Kneser-Ney backoff technique.**

What is the continuation probability of *"is"* ?

a) 0.0078
b) 0.0076
c) 0.0307
d) 0.0081

Answer: b

Solution: Refer week 3 lecture 12

Continuation probability of is = 1/130 = 0.0076

The numerator means the number of different string types preceding the final word, (here only 1 type– language processing is) and the denominator means the number of different possible n-gram types , in this case trigram = 130

Question 5: What will be the value of P(is| language processing) using Kneser-Ney backoff technique and choose the correct answer below. . Please follow the paragraph in Question .

a) 0.5
b) 0.6
c) 0.8
d) 0.7

Answer: c

Solution: Refer week 3 lecture 12

P(is| language processing) = ⅘ + 0*0.0076 = 0.8 [as d=0 so lambda = 0]

In this example is equal to the frequency of language processing *: the frequency of language processing is (here it occurs 4 times) plus the frequency of language processing can (occurs only once). Therefore, for word is, firstTerm(is) = 4/(4+1) = 0.8

Question 6. What is the value of P(can| language processing)? Please follow the paragraph in Question 5

a) 0.1
b) 0.02
c) 0.3
d) 0.2

Answer: d

Solution: Refer week 3 lecture 12

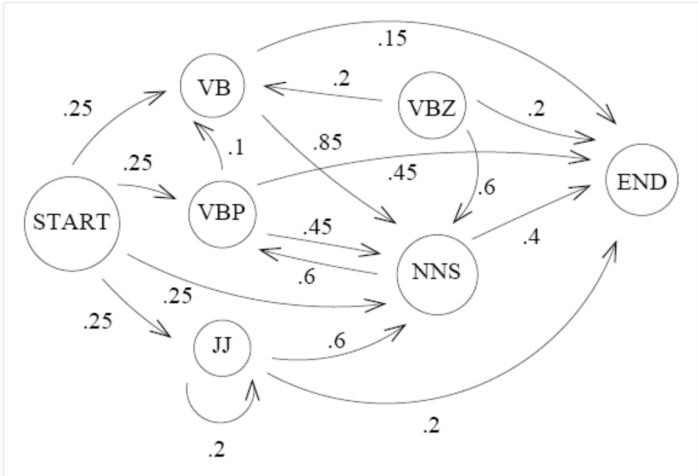
Similarly P(can| language processing) = ⅕ + 0*Continuation probability = 0.2

Language processing * occurs 5 times, language processing can occur only once.

Question 7: Consider the HMM given below to solve the sequence labeling problem of POS tagging. With that HMM, calculate the probability that the sequence of words "bright lights" will be assigned the following parts of speech;

VB NNS

	bright	*lights*
JJ	**0.00158**	**0**
NNS	**0**	**0.00050**
VB	**0.00125**	**0**
VBP	**0.00081**	**0**
VBZ	**0**	**0.00005**



The above table contains emission probability and the figure contains transition probability

a) $4.80 * 10^{-8}$
b) $5.31 * 10^{-8}$

c) $5.31 * 10^{-7}$
d) $4.96 * 10^{-8}$

Answer: b

Solution:

P(bright lights, VB NNS)
= P(VB | start) × P(bright | VB) × P(NNS | VB) × P(lights | NNS) × P(end | NNS)

= 0.25 × 0.00125 × 0.85 × 0.00050 × 0.40

= 5.31×10^{-8}